



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHREE MUDDEERESWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

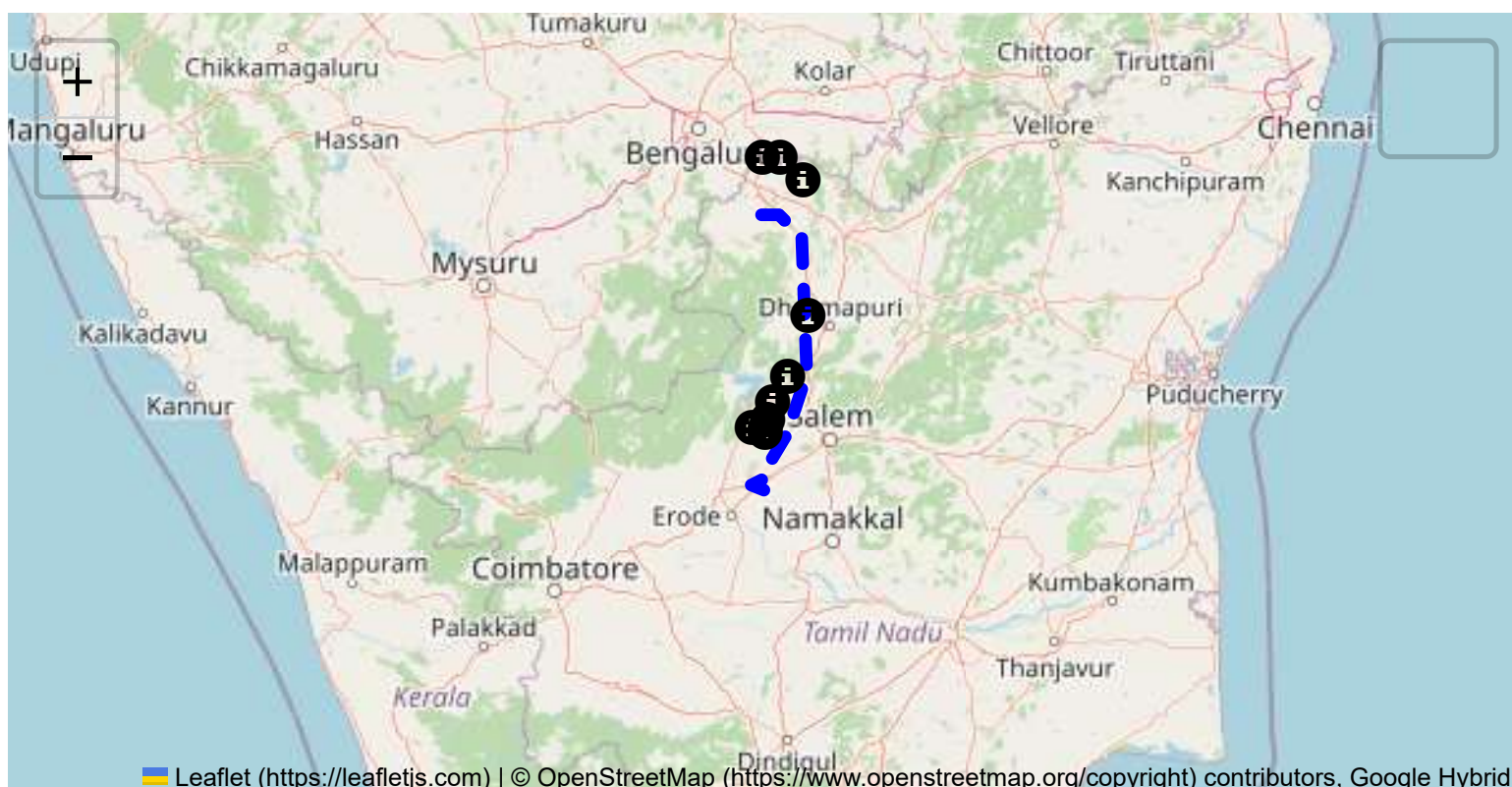
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

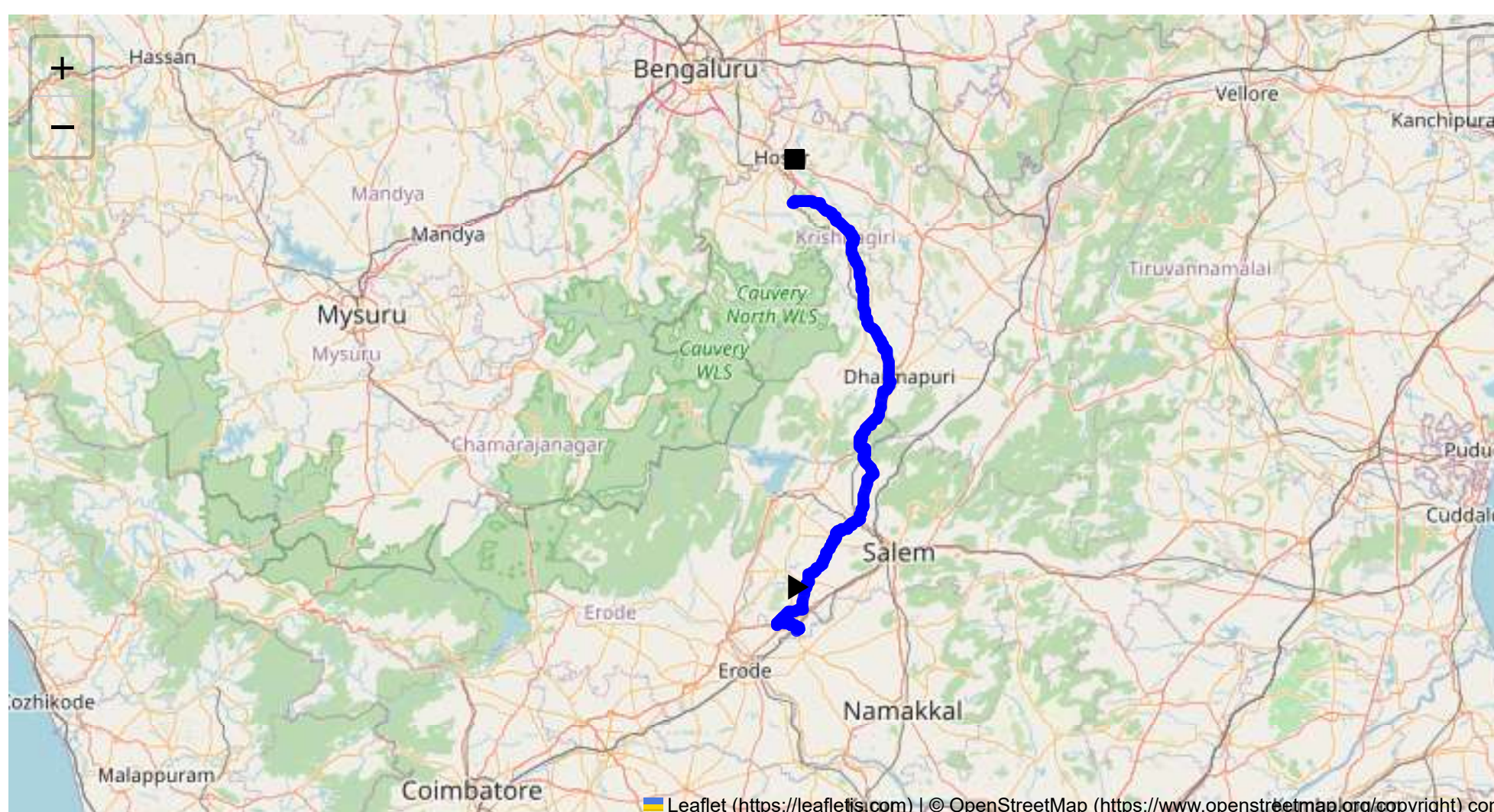
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 173.44 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.4 hours
Start: (11.4381, 77.8734)
End: (12.607314, 77.861046)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 173.44 kilometers across Tamil Nadu, starting from CVQF+23W, Sangagiri to JV46+X97, Kelamangalam, traveling through prominent locations such as Padaiveedu, Desavilakku, Thoppur, and Rayakottai. It comprises a combination of state highways and local roads, which are frequently used by both passenger and commercial vehicles.

Weather Conditions

The weather in Tamil Nadu is predominantly tropical, with high temperatures and humidity levels for most of the year. During the monsoon season (June to September), heavy rainfall can lead to waterlogging and slippery roads, especially in low-lying areas along the route. In winter months (November to February), fog is occasionally a concern in the early mornings, reducing visibility.

Traffic Patterns

- **Peak Hours:** Morning (8 AM - 10 AM) and Evening (5 PM - 8 PM) are typically the busiest times.
- **Congestion-Prone Areas:** Key congested points include the Salem - Kochi Hwy and areas around Thoppur due to their significance as major transit routes.

Road Quality and Infrastructure

Road quality varies along the route. Highways are relatively well-maintained, but certain local roads may have potholes and require caution. Infrastructure, such as signage and lighting, is generally sufficient on main highways but may be lacking in rural areas.

Alternative Routes

In case of emergencies, alternative routes could involve taking regional bypasses or detours through neighboring towns like Tiruchengode and Anthiyur. These alternatives, though longer, may provide less congested and safer pathways during critical situations.

Regulatory Considerations

Tamil Nadu has specific regulations regarding the transport of hazardous materials, including designated routes and times for such goods. Night travel often requires special permits.

Historical Incidents

While the area is usually safe, there have been sporadic reports of accidents involving heavy vehicles, primarily attributed to adverse weather conditions or driver fatigue.

Environmental Considerations

The route passes near several environmentally sensitive areas, such as agricultural zones and water bodies, necessitating cautious navigation to prevent potential spillage or contamination.

Communication Coverage

Most of the route has good cellular coverage, although there may be dead zones in less-populated stretches, particularly between Desavilakku and Thoppur.

Emergency Response Times

Emergency services are fairly responsive around urban centers like Salem and Dharmapuri, with estimated response times of 30-45 minutes. Response times may be longer in rural segments.

Overall Risk Assessment

This route poses moderate risk to heavy vehicle travel. Key concerns include weather-related hazards, specific congestion points, and the condition of secondary roads. However, with careful planning around peak traffic times and weather forecasts, alongside adherence to regulatory measures, risks can be effectively mitigated. Ensuring vehicles are equipped with reliable communication tools and emergency kits is advisable given the occasional signal shortages and variable emergency response times.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45545, 77.81812	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56352, 77.90078	30 KM/Hr
11	Turn	Medium	11.59189, 77.90645	30 KM/Hr
12	Turn	Medium	11.59328, 77.90809	30 KM/Hr
13	Turn	Medium	11.63867, 77.94841	30 KM/Hr
14	Turn	High	11.64217, 77.95200	15 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	Medium	11.67858, 77.96834	30 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72504, 78.01680	30 KM/Hr
21	Turn	Medium	11.73766, 78.03445	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74265, 78.04710	30 KM/Hr
24	Turn	Medium	11.78005, 78.05788	30 KM/Hr
25	Turn	Medium	11.86928, 78.08384	30 KM/Hr
26	Turn	Medium	11.88009, 78.08164	30 KM/Hr
27	Turn	Medium	11.91277, 78.05853	30 KM/Hr
28	Turn	Medium	11.92667, 78.06517	30 KM/Hr
29	Turn	Medium	11.93213, 78.06468	30 KM/Hr
30	Turn	Medium	11.93480, 78.06255	30 KM/Hr
31	Turn	Medium	11.93746, 78.05331	30 KM/Hr
32	Turn	Medium	11.95531, 78.04698	30 KM/Hr
33	Turn	High	11.95854, 78.04990	15 KM/Hr
34	Turn	High	11.96015, 78.04945	15 KM/Hr
35	Turn	Medium	11.96266, 78.05115	30 KM/Hr
36	Turn	High	11.96433, 78.05113	15 KM/Hr
37	Turn	High	11.96542, 78.04938	15 KM/Hr
38	Turn	High	11.96650, 78.04927	15 KM/Hr
39	Turn	Medium	12.12036, 78.13284	30 KM/Hr
40	Blind Spot	Blind Spot	12.51393, 78.03186	10 KM/Hr
41	Turn	Medium	12.51379, 78.02933	30 KM/Hr
42	Turn	Medium	12.51624, 78.02577	30 KM/Hr
43	Turn	Medium	12.52904, 78.02046	30 KM/Hr
44	Turn	High	12.60668, 77.93608	15 KM/Hr
45	Turn	Medium	12.61380, 77.90329	30 KM/Hr
46	Turn	Medium	12.61313, 77.90179	30 KM/Hr
47	Turn	Medium	12.61415, 77.87192	30 KM/Hr
48	Turn	High	12.61115, 77.86692	15 KM/Hr
49	Blind Spot	Blind Spot	12.61214, 77.86546	10 KM/Hr
50	Turn	High	12.61135, 77.86573	15 KM/Hr
51	Turn	Medium	12.61004, 77.86507	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
32	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
33	college	Government Polytechnic College	12.613737, 77.8786607	30 km/h	Medium
34	school	Government Model Higher Secondary School	12.615863, 77.8787172	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56352, 77.90078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74265, 78.04710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78005, 78.05788



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86928, 78.08384



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88009, 78.08164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91277, 78.05853



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92667, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93213, 78.06468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93480, 78.06255



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93746, 78.05331



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96433, 78.05113



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96542, 78.04938



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12036, 78.13284



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.51393, 78.03186



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51379, 78.02933



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51624, 78.02577



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52904, 78.02046



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.60668, 77.93608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.61380, 77.90329



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.61313, 77.90179



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.61415, 77.87192
